There is also a considerable difference in the reaction time of the different interactions. The strong interaction manifests itself in its ability to react in very short times. For example, for a particle, which passes an atomic nucleus of about 10^{-15} m in diameter with a velocity of approximately 10^{8} m s^{-1} (i.e. with a kinetic energy of ~50 MeV for a proton and 0.03 MeV for an electron), the time of strong interaction is about 10^{-23} s. This is about the time of rotation of the atomic nucleus. The weak interaction force requires a much longer reaction time and explains why leptons such as electrons and photons do not react with atomic nuclei but do react with the electron cloud of the atom, which has a diameter on the order of 10^{-10} m. There is sufficient time in passing this larger diameter for the weak interaction force to be effective. It is expected that the electromagnetic force and the gravitational force are very fast compared to the weak nuclear force.

2.3. WAVES AND PARTICLES

It is daily experience that moving bodies have a kinetic energy, E_v, which involves a mass and a velocity, v. Less familiar is the concept of Planck (1900) in which light moves in wave packets of energy:

$$E = \mathbf{h}\nu \tag{2.3}$$

Here ν is the frequency of light with wavelength λ

$$\mathbf{c} = \nu\lambda \tag{2.4}$$

and **h** the Planck constant, 6.63×10^{-34} J s.

Einstein — in the theory of the photoelectric effect —, and Compton — in the theory of the scattering of photons (Ch. 6) — showed that photons have not only a discrete energy, but also a discrete momentum (p_v).

$$p_v = E_v/\mathbf{c} \tag{2.5}$$

Photons seem to collide with other particles as if they have a real mass and velocity as in the classical mechanical expression for momentum: $p = mv$. If we put $v = \mathbf{c}$ and equate with (2.5) we obtain a relativistic mass of the photon as

$$m_v = E_v/\mathbf{c}^2 \tag{2.6}$$

This is the mass-energy relation of Einstein.

There are many examples of mass properties of photons. To the two mentioned above we may add the solar pressure (i.e. photons from the sun which push atoms away from the sun and into space), which has played a significant part in the formation of our planetary system, and measurements showing that photons are attracted by large masses through the gravitational force. Thus we see the evidence for the statement in the beginning that all moving elementary particles must have relativistic mass, even if the rest mass is zero.

It is reasonable to assume, as de Broglie did in 1924, that since photons can behave as moving particles moving particles may show wave properties. From the previous equations, we can — by replacing **c** by v — derive that the wavelength of such *matter waves* is

$$\lambda = \mathbf{h}/mv \tag{2.7}$$

This relation is of importance in explaining nuclear reactions, and has led to practical consequences in the use of electron diffraction, and in the development of electron microscopy.

The wave and particle properties of matter complement each other (the *complementarity principle*; N. Bohr, 1928). Throughout this book we either use models based on wave properties or sometimes on particle properties, depending on which more directly explain the particular phenomenon under discussion.

2.4. FORMATION AND PROPERTIES OF SOME ELEMENTARY PARTICLES

The track formed by a moving particle in the magnetic field of a bubble chamber is characterized by its width, length and curvature. From a kinematic analyses of the tracks it is possible to determine the mass and charge of the particles involved, see Ch. 3, and Ch. 9. Further, as seen at points A-B and E-F in Figure 2.2, the interruption of a track can indicate formation of an uncharged particle, as they do not form visible tracks. From a knowledge of the tracks formed by known particles, "strange tracks" can be analyzed to identify new particles in bubble chamber pictures and to assign their properties. All these properties have to be quantized, so new quantum states have been introduced, like baryon number, statistics, symmetry, parity, hypercharge, isospin, strangeness, color, etc. in addition to spin. It is found that many of the particles observed are sensitive to only one or two of the forces of nature (§2.2), which serves as an additional aid in their classification. Nevertheless, the array of properties assigned to the hundreds of "elementary particles" which have been discovered resembles the situation when 40 radioelements were reported between uranium and lead, as described in chapter 1.

Before we proceed to the order which has evolved from this picture we must describe some of the concepts used to define these particles. It has been practical to divide the particles according to their masses: (i) *baryons* are the "heavy" ones (protons, neutrons, hyperons and nuclei), (ii) *leptons* are the "light" ones (the electron, the neutrino and the muon); (iii) *mesons* have "intermediate" masses; these include the π-meson, the K-meson, etc. The baryons and the mesons have also been considered as *hadrons*, "hard" or "strong" particles as they take part in the strong nuclear force. Such properties were used to develop a table of elementary particles, Table 2.2.